Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

AI Engineer and Forward Deployed Builder with 8+ years shipping production systems inside enterprise customer environments. Built **AgentProbe** - an open-source agentic testing toolkit running three LangGraph agents (Orchestrator, RAG Tracer, Auto Fix), a 5-metric LLM-as-judge eval engine, and a 15-endpoint FastAPI backend directly on the **Anthropic Claude API**. Shipped 20+ production modules at scale (1M+ records, 100K+ MAU) across 120+ Agile sprints at Entrata; delivered **AI on FHIR** - a HIPAA-compliant NLP system on structured clinical records. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). Targeting roles where I can embed with customers, build production LLM applications, and harden agent systems for enterprise use. Case studies: AgentProbe | AI on FHIR

CORE COMPETENCIES

LLM Applications (Anthropic API) | Agent Orchestration (LangGraph) | Prompt Engineering | MCP Architecture / Tool Endpoints | RAG Pipelines & Evals | Customer-Embedded Delivery | Healthcare AI (HIPAA/FHIR) | Enterprise Python at Scale | FastAPI / Microservices | Multi-Agent Systems | LLM-as-Judge Evaluation | Production AI Deployment

EXPERIENCE

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Virginia, USA

- Improved student performance in Python and database courses by delivering targeted mentoring across lectures, labs, and office hours for 30+ undergraduates.
- Built supplementary Python exercises for iterative sprint-style delivery, accelerating concept reinforcement cycles.

George Mason University

Aug 2024 - Aug 2025

Research Associate - Applied ML - Virginia, USA

- Built regression models ($R^2 = 0.977$) in Python and R across WHO and UK Road Safety datasets spanning 190+ countries - multi-source data collection, transformation, validation, and stakeholder reporting.
- Deployed Python anomaly detection auditor (Isolation Forest) across 3 research pipelines - cut manual QA time by 70% and surfaced anomalies that manual review missed.
- Designed Tableau, ggplot2, and Quarto dashboards for faculty leadership - improved research comprehension across two cohorts by 40%.

Entrata India (Db Xento Systems)

Oct 2018 - Aug 2024

Software Engineer - Enterprise Platform - Pune, India

- **Delivered 20+ production modules embedded in enterprise customer environments** across 120+ Agile sprints (Python, PostgreSQL, JavaScript, RabbitMQ) for 100K+ monthly active users - 99.5% uptime sustained over 6+ consecutive months.
- Cut microservices deployment times by 40% by leading Docker-based containerization for 30%+ of modules; improved production query performance by 30% by tuning 200+ SQL queries across all client environments.
- **Deployed Python + Anthropic/OpenAI pipelines on 1.6M financial records** with output validation and quality gates - production AI on regulated data with zero compliance incidents.
- Achieved 100% PHI compliance by migrating all client data to AWS S3 with Libsodium-encrypted pipelines; built 5+ internal tools cutting resolution time by 40%.
- Eliminated 80% of manual operational workload by designing and automating decision support workflows using Python and PL/SQL.
- Conducted 300+ peer code reviews; mentored 5+ engineers on secure coding practices and TDD/BDD standards.

Entrata India

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer - Pune, India

- Built and optimized 20+ full-stack property management modules (Python, PostgreSQL) - improved system stability by 35%, 99.5% uptime, 25% faster report generation.
- Resolved 100+ high-priority bugs; delivered production-ready APIs and UI modules from day one using Git-based workflows and TDD practices.

PROJECTS

AgentProbe: Agentic AI Testing Toolkit

LIVE - ANTHROPIC API

Python, LangGraph, FastAPI, ChromaDB, Anthropic Claude API, OpenAI, Groq, Ollama, React, Docker, GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

Built a 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with tool calling, inter-agent state handoffs, and multi-provider LLM routing - **Anthropic Claude as primary provider**. Deployed as a 15-endpoint FastAPI backend exposing tool endpoints (MCP-compatible pattern) running 100% locally at zero cost. Automated quality gates with a 5-metric LLM-as-judge eval engine (factual accuracy, hallucination, retrieval, completeness, compliance) integrated into CI/CD - failing runs blocked automatically before production. Delivered end-to-end observability via React dashboard with trace-level agent monitoring and regression drift detection across concurrent test suites.

AI on FHIR: NLP-Driven Medical Records Query System

HEALTHCARE AI

Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Reduced manual clinical retrieval time by ~60% by building a Flask + React NLP platform translating natural language queries (EN/ES) to FHIR APIs using LLMs, BERT, and embeddings. **HIPAA-compliant RBAC** across structured medical records - production healthcare AI with compliance guardrails built in from day one.

CrashClarity: Large-Scale Road Accident Analysis (104K Records)

ML PIPELINE

Python, R, scikit-learn, Random Forest, Logistic Regression, ggplot2, Feature Engineering, GitHub Pages

Achieved 85%+ accuracy on 104K+ UK accident records by building a Random Forest + Logistic Regression pipeline with feature engineering across weather, road, and time variables. Deployed interactive dashboard on GitHub Pages for non-technical stakeholder consumption.

EDUCATION

M.S. Data Analytics Engineering - George Mason University — 2024 - 2026

GPA: 3.87 | Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering — 2013 - 2017

India

TECHNICAL SKILLS

AI & Agents: LangGraph, LangChain, Multi-Agent Orchestration, Tool Calls, MCP Architecture, Anthropic API, OpenAI API, Groq, Ollama, RAG Pipelines, ChromaDB, Embeddings, LLM-as-Judge, Eval Frameworks, Agent Observability, Prompt Engineering

Languages: Python, JavaScript, TypeScript, SQL, PHP, R, Bash, HTML/CSS

Backend & Cloud: FastAPI, Flask, Microservices, RabbitMQ, AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions)

ML & Data: PyTorch, TensorFlow, XGBoost, Scikit-learn, Isolation Forest, Random Forest, BERT, spaCy, Hugging Face, Pandas, NumPy, Feature Engineering

Databases: PostgreSQL, MySQL, Oracle, NoSQL - schema design, query tuning, indexing, GraphQL